

Technical Assignment – Recommendation System

System Challenge

Objective

Design and build an **intelligent recommendation system from scratch**.

This assignment is intended to evaluate your **problem-solving skills, engineering ability, product thinking, and understanding of modern AI/ML systems**.

There is no single correct solution. We are interested in **how you think, design, implement, evaluate, and communicate your ideas**.

1. Working Recommendation System

Build a functional recommendation engine for a **use case of your choice**.

You are free to choose:

- Dataset
- Recommendation technique
- Technology stack
- Architecture
- Evaluation methodology

Examples include, but are not limited to:

- E-commerce recommendations
- Movie recommendations
- Music recommendations
- Food recommendations
- Fashion recommendations
- Beauty recommendations
- Book recommendations
- Any innovative recommendation use case

The system should be functional and demonstrate a clear recommendation strategy.

2. Testing Interface — Mandatory

Create a simple **deployed UI** that allows evaluators to interact with your recommendation system.

The interface should allow us to:

- Enter or select inputs
- Generate recommendations
- Clearly observe recommendation outputs
- Understand why or how recommendations are being generated, where applicable

The focus is on **functionality and usability rather than visual perfection**.

3. Documentation — Mandatory

Provide proper documentation explaining your solution.

Your documentation should cover:

- Problem statement
- Use case and motivation
- Approach
- System architecture
- Recommendation methodology
- Dataset selection
- Technologies used
- Assumptions made
- Key design decisions
- Evaluation methodology
- Test cases
- Known limitations
- Future improvements

The documentation should be clear enough for another engineer to **understand the system and reproduce it**.

Test Cases

Include representative examples demonstrating both:

Successful Scenarios

Examples where your recommendation system performs well.

Failure Scenarios

Cases where the system struggles or produces imperfect recommendations.

Understanding and clearly explaining the limitations of your system is considered a **strength**.

Evaluation Metrics

Define how the success of your recommendation system should be measured.

Depending on your approach, you may consider metrics such as:

- Precision
- Recall
- NDCG
- MAP
- Diversity
- Coverage
- Novelty
- Latency
- User Satisfaction
- Business Metrics

If standard recommendation metrics are not applicable to your approach, clearly explain and justify the evaluation methodology you have chosen.

What We Will Evaluate

Your submission will be assessed across the following dimensions:

- Problem-solving approach
- Technical implementation

- Recommendation quality
 - AI/ML understanding
 - System design
 - Product thinking
 - Code quality
 - Documentation
 - Creativity
 - User experience
 - Engineering best practices
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Bonus Challenge — Highly Recommended

If your recommendation system is inspired by an existing product or platform such as **Netflix, Spotify, Amazon, Instagram, YouTube, Nykaa, or similar platforms**, consider designing your UI to take inspiration from that product.

Additionally, include a short comparison covering:

- Similarities
- Differences
- Current limitations
- Areas for improvement
- What you would build next with more time

This will help us understand your **product thinking, ability to benchmark against existing solutions, and understanding of real-world recommendation systems**.

Submission Requirements

Your submission must include **all three** of the following:

1. GitHub Repository

Provide a link to a public or accessible GitHub repository containing:

- Source code
- Configuration files
- Sample dataset, where applicable
- Setup instructions

- Any other required project files

2. Deployment Link

Provide a **working deployment link** where we can directly test your recommendation system.

The deployed application should be accessible to evaluators without requiring local setup, wherever possible.

3. Proper Documentation

Provide clear documentation explaining:

- What you built
- How it works
- How to run it
- Architecture
- Recommendation approach
- Dataset
- Evaluation
- Test cases
- Limitations
- Future improvements

Your documentation may be provided through the GitHub repository, a dedicated documentation file, or another accessible format.

Final Note

This assignment is intentionally **open-ended**.

We value **thoughtful engineering, originality, clarity of execution, and sound technical decisions** over building the most complex solution.

You are encouraged to showcase your:

Creativity → Curiosity → Product Thinking → Engineering → Execution

We look forward to seeing what you build.